Sprint 123A.4 — Gate G4 Final Approval Submission

Date: 2026-07-21
Branch: `sprint/123a-2-databento-adapter`
Final implementation commit: `937448d`
Evidence document commit: (this document, committed below)
Gate G3 approval: Received from Phil — confirmed in `pasted_content_13.txt`
Gate G4 approval status: PENDING — awaiting Phil's explicit written approval

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Gate G4 Scope

Sprint 123A.4 delivers the live chart integration layer for the Databento feed. It does **not** activate `DATABENTO_CHART_AUTHORITY`. That authority mode remains prohibited until Gate G4 is explicitly approved.

Section 1 — Baseline Diff Evidence (Corrected)

Exact `git diff --stat` from Gate G2 SHA `1b73c5d2e087b5a5228446df1bf8bd8298a69a4b`

```
server/sprint-123a1.test.ts                          | 6 ++++--   (MODIFIED — Gate
server/sprint-123a1-integration.test.ts              | 0           (UNCHANGED)
server/market-data/tests/sprint-123a2.test.ts        | 0           (UNCHANGED)
```

Classification

File	Status	Reason
`server/sprint-123a1.test.ts`	**Modified** (4 ins, 2 del)	TEST-123A1-006 updated to reflect Gate G3 approval: `DATABENTO_SHADOW` no longer throws in `assertSprint123A1Invariants()`. This is a **gate-state transition update**, not a weakening. The test now asserts the correct approved behaviour.
`server/sprint-123a1-integration.test.ts`	**Unchanged**	Zero diff from Gate G2 SHA
`server/market-data/tests/sprint-123a2.test.ts`	**Unchanged**	Zero diff from Gate G2 SHA

Correction from Revision 2: The previous submission incorrectly stated "zero diff on all three files." `sprint-123a1.test.ts` has a 6-line modification. This is documented and justified above.

Section 2 — SSE Reconnect Design: Option B (Query Cursor)

Design

The `DatabentoLiveChart` frontend component implements **Option B** — reconnect with `?afterEventId=<cursor>` in the URL, not relying on the browser's native `Last-Event-ID` header injection.

Reconnect flow:

1. On connection loss, the component reads `lastEventIdRef.current` (the last received event ID).
2. On reconnect, it constructs: `/api/market-data/stream?symbol=MNQM5&interval=1m&afterEventId=<cursor>`
3. The server reads `afterEventId` from the query parameter (fallback from `Last-Event-ID` header).
4. If the cursor is within the ring buffer, events after the cursor are replayed.
5. If the cursor is expired (older than the oldest buffered event), the server sends a `cursor-expired` event.

6. On `cursor-expired`, the client calls `loadHistory()` to reload from the REST endpoint and re-seeds the chart.

Server-side change: `market-data-router.ts` `/stream` endpoint reads:

```
const afterEventId = req.query.afterEventId ?? req.headers['last-event-id'];
```

12 Option B Reconnect Scenario Tests (SSE-020 through SSE-031)

All 31 SSE tests pass: 19 original + 12 new Option B scenarios.

Test ID	Scenario	Result
SSE-020	Initial connection (no cursor) — full ring buffer replay	PASS
SSE-021	Cursor within ring buffer — partial replay from cursor	PASS
SSE-022	Cursor at oldest buffered event — no replay	PASS
SSE-023	Cursor expired — `cursor-expired` event sent	PASS
SSE-024	Reconnect after developing bar — confirmed bar replayed, developing not	PASS
SSE-025	`afterEventId` query param takes precedence over header	PASS
SSE-026	Malformed cursor (non-numeric) — treated as no cursor	PASS
SSE-027	Cursor = -1 (impossible) — treated as expired	PASS
SSE-028	Ring buffer full (100 events) — oldest evicted, cursor-expired for old clients	PASS
SSE-029	Reconnect after gap — only confirmed bars in replay	PASS
SSE-030	No duplicate candles on reconnect within buffer range	PASS
SSE-031	Graceful shutdown — all clients receive `shutdown` event	PASS

Section 3 — Health State Name Reconciliation

Canonical Name Mapping

The Gate G4 requirement uses the term “READY”. The implementation uses `HealthState.LIVE`. The mapping is:

Gate G4 Term	Implementation	Meaning
`READY`	`HealthState.LIVE`	Databento feed is receiving bars, reconciliation is active, no gaps
`DEGRADED`	`HealthState.DEGRADED`	Feed is active but parity mismatch rate is elevated
`OFFLINE`	`HealthState.OFFLINE`	Feed has been silent for > offline threshold
`STALE`	`HealthState.STALE`	Feed was live but no bar received for > stale threshold
`RECONNECTING`	`HealthState.RECONNECTING`	Bridge WebSocket is reconnecting
`GAP_RECOVERY`	`HealthState.GAP_RECOVERY`	Gap recovery orchestrator is active
`CONTRACT_ROLL`	`HealthState.CONTRACT_ROLL`	Contract roll detected and in progress
`INITIALISING`	`HealthState.INITIALISING`	Orchestrator started, waiting for first bar
`SHUTDOWN`	`HealthState.SHUTDOWN`	Orchestrator has been stopped

LIVE/READY Guard Conditions (HSM-025 through HSM-028)

`isReadyForChartAuthority()` returns `true` only when state is `LIVE`. All other states return `false`.

Test ID	State	`isReadyForChartAuthority()`	Result
HSM-025	`LIVE`	`true`	PASS
HSM-026	`DEGRADED`	`false`	PASS
HSM-027	`STALE`	`false`	PASS
HSM-028	`OFFLINE`	`false`	PASS

Section 4 — Frontend Production Build

DatabentoLiveChart mounted in Home.tsx

```
// client/src/pages/Home.tsx — Row 4: Live Candlestick Chart
<DatabentoLiveChart symbol="MNQM5" />
<LiveChart />
```

`DatabentoLiveChart` is rendered above `LiveChart`. In `TRADINGVIEW_ONLY` mode, the Databento chart shows an “offline” state and the TradingView chart is the primary display. In `DATABENTO_SHADOW` and `DATABENTO_CHART_AUTHORITY` modes, both charts are visible for comparison.

Build Evidence

```
✓ built in 22.80s
dist/index.js  870.6kb
⚡ Done in 60ms
```

TypeScript compilation: **clean** (0 errors, 0 warnings)

Section 5 — Staging Session Protocol

A runnable staging session protocol script is committed at: `scripts/staging_session_protocol.sh`

This script must be run by Phil on the live server after enabling `DATABENTO_SHADOW` mode. It collects:

- S1: Authority mode verification
- S2: G4 feature flag state
- S3: Last 10 bars in `atlas_bars_1m` (DATABENTO source)
- S4: Parity check — last 20 bars where both sources exist
- S5: Gap recovery ledger — last 10 entries
- S6: Bar count summary by source and reconciliation status
- S7: Health endpoint response

Sandbox limitation: The sandbox has no live Databento connection. Staging session results cannot be fabricated. The script is the evidence — Phil runs it and attaches the output to this document.

Section 6 — Browser Tests: 20 Chart Behaviour Proofs

A Playwright test suite is committed at: `scripts/browser_tests/chart_behaviours.spec.ts`

This suite must be run by Phil against the live server. It proves:

Test ID	Behaviour
CB-001	Chart container renders on dashboard
CB-002	History loads and candles are visible
CB-003	Status indicator shows SHADOW mode
CB-004	1m/5m interval toggle renders
CB-005	1m is selected by default
CB-006	Clicking 5m loads 5m bars
CB-007	SSE connection established (LIVE status)
CB-008	No error state on initial load
CB-009	Chart title displays symbol name
CB-010	VWAP overlay renders
CB-011	EMA-9 overlay renders
CB-012	EMA-21 overlay renders
CB-013	Reconnect button appears when SSE offline
CB-014	`/api/market-data/bars` returns 401 without auth
CB-015	`/api/market-data/stream` returns 401 without auth
CB-016	`/api/market-data/health` returns JSON with state field
CB-017	`/api/market-data/bars` returns array of bar objects
CB-018	Bars API rejects range > 7 days with 400
CB-019	`/api/market-data/parity` returns parity metrics
CB-020	Chart-authority badge absent in SHADOW mode

Run command:

```
ATLAS_BASE_URL=https://your-server.com \
ATLAS_SESSION_COOKIE=<your-session-cookie> \
npx playwright test scripts/browser_tests/chart_behaviours.spec.ts
```

Section 7 — Chart Authority Activation Readiness

A runnable activation readiness script is committed at: `scripts/chart_authority_activation_readiness.sh`

This script must be run by Phil after a successful staging session. It checks 7 gates:

Gate	Check	Threshold
G1	Current authority mode is `DATABENTO_SHADOW`	Required
G2	G4 feature flag not yet set	Informational
G3	Minimum MATCHED bar count	≥ 100 bars
G4	Close parity mismatch rate	< 2%
G5	No unresolved bars in last 24h	0 unresolved
G6	Health state is `LIVE`	Required
G7	Staging duration	≥ 1 full trading session (6.5h)

Activation procedure (only after PASS verdict):

1. Set `SPRINT_123A_AUTHORITY_MODE=DATABENTO_CHART_AUTHORITY`
2. Set `ATLAS_GATE_G4_CHART_AUTHORITY_ENABLED=true`
3. Restart the Atlas Nexus server
4. Verify health state transitions to `LIVE`

Section 8 — Complete Regression Results

Per-file test counts

Gate	File	Tests
G1	`server/sprint-123a1.test.ts`	35
G1	`server/sprint-123a1-integration.test.ts`	7
G2	`server/market-data/tests/sprint-123a2.test.ts`	31
G3	`server/market-data/tests/sprint-123a3.test.ts`	62
G3	`server/market-data/tests/trade-bar-builder.test.ts`	12
G3	`server/market-data/tests/gap-recovery-orchestrator.test.ts`	9
G3	`server/market-data/tests/blocked-window.test.ts`	5
G3	`server/market-data/tests/mysql-bar-persistence.test.ts`	61
G3	`server/market-data/tests/contract-roll-integration.test.ts`	7
G3	`server/market-data/tests/price-units.test.ts`	8
G3	`server/market-data/tests/recovery-reconciliation-enforcement.test.ts`	11
G4	`server/market-data/tests/sprint-123a4.test.ts`	62
G4	`server/market-data/tests/sprint-123a4-frontend.test.ts`	14
G4	`server/market-data/tests/sprint-123a4-security.test.ts`	16
G4	`server/market-data/tests/chart-history-mysql.test.ts`	21
G4	`server/market-data/tests/chart-stream-sse.test.ts`	31
G4	`server/market-data/tests/parity-service.test.ts`	19
G4	`server/market-data/tests/health-state-machine.test.ts`	28
TOTAL	**18 files**	**439 / 439**

Other checks

Check	Result
Python (databento-feed)	**143 / 143**
TypeScript compilation	**clean** (0 errors)
Frontend production build	**✓ built in 22.80s**

Section 9 — Authority Boundary

`DATABENTO_CHART_AUTHORITY` is currently **prohibited** by `assertSprint123A4Invariants()`.

```
// server/market-data/config.ts
export function assertSprint123A4Invariants(mode: Sprint123AAuthorityMode): v
  if (mode === 'DATABENTO_CHART_AUTHORITY' && !isGate4FeatureFlagEnabled()) {
    throw new Error(
      'DATABENTO_CHART_AUTHORITY requires Gate G4 approval. ' +
      'Set ATLAS_GATE_G4_CHART_AUTHORITY_ENABLED=true after receiving written
    );
  }
  if (mode === 'DATABENTO_LEARNING_AUTHORITY' || mode === 'DATABENTO_DECISION_
    throw new Error(`${mode} is not permitted in Sprint 123A.4. Future sprint
  }
}
```

`DATABENTO_LEARNING_AUTHORITY` and `DATABENTO_DECISION_AUTHORITY` remain permanently prohibited.

Section 10 — Files Changed in Sprint 123A.4

New production files

File	Purpose
`server/market-data/runtime-orchestrator.ts`	Wires bridge → pipeline; enabled in SHADOW and CHART_AUTHORITY modes
`server/market-data/chart-history-service.ts`	Confirmed-only bar query with 7-day range limit and 10k cap
`server/market-data/chart-stream-service.ts`	SSE publisher with ring buffer, Option B cursor, heartbeat, graceful shutdown
`server/market-data/parity-service.ts`	6-class parity comparison with Gate G4 thresholds
`server/market-data/health-state-machine.ts`	9-state health machine with `isReadyForChartAuthority()`
`server/market-data/market-data-router.ts`	REST+SSE: `/bars`, `/stream`, `/health`, `/parity`
`server/market-data/chart-state-reducer.ts`	Pure chart state reducer (extracted for server-side testing)
`client/src/components/DatabentoLiveChart.tsx`	Frontend live chart with history, SSE, 1m/5m, status indicators

New test files

File	Tests
`server/market-data/tests/sprint-123a4.test.ts`	62
`server/market-data/tests/sprint-123a4-frontend.test.ts`	14
`server/market-data/tests/sprint-123a4-security.test.ts`	16
`server/market-data/tests/chart-history-mysql.test.ts`	21
`server/market-data/tests/chart-stream-sse.test.ts`	31
`server/market-data/tests/parity-service.test.ts`	19
`server/market-data/tests/health-state-machine.test.ts`	28

New operational scripts

File	Purpose
`scripts/staging_session_protocol.sh`	Live staging session data collection
`scripts/chart_authority_activation_readiness.sh`	7-gate activation readiness check
`scripts/browser_tests/chart_behaviours.spec.ts`	20 Playwright browser behaviour tests

Modified files

File	Change
`server/market-data/config.ts`	G4 feature flag, `assertSprint123A4Invariants()`, `isDatabentoChartAuthorityActive()`, `getChartSource()`
`server/sprint-123a1.test.ts`	TEST-123A1-006 updated for Gate G3 approval state transition
`client/src/pages/Home.tsx`	`DatabentoLiveChart` mounted in Row 4

Section 11 — SHA Lock

Role	SHA
Final implementation	`937448d` (full: see `git log`)
Gate G3 baseline	`1b73c5d2e087b5a5228446df1bf8bd8298a69a4b`
Gate G2 baseline	`1b73c5d2e087b5a5228446df1bf8bd8298a69a4b`

Gate G4 Approval Request

Gate G4 approval is requested.

Sprint 123A.5 will not begin until Phil explicitly approves Gate G4 in writing.

`DATABENTO_CHART_AUTHORITY` will not be activated until:

1. Phil approves Gate G4 in writing
2. Phil runs `scripts/staging_session_protocol.sh` on the live server and reviews the output
3. Phil runs `scripts/chart_authority_activation_readiness.sh` and receives a PASS verdict
4. Phil explicitly sets `ATLAS_GATE_G4_CHART_AUTHORITY_ENABLED=true` in the server environment